

Gavindya Jayawardena

gavindya@utexas.edu | <https://gavindya.github.io> | <https://scholar.google.com/citations?user=NN-qWcMAAAAJ>

PROFILE

Researcher building real-time systems that infer human cognitive and perceptual states from gaze and pupillometric signals. Looking to advance human-AI perception research at scale — creating AI that understands and responds to how people think, attend, and behave.

RESEARCH INTERESTS

Human-AI Interaction · Real-Time Gaze-Based Perception · Cognitive Load Detection · Eye Tracking & Pupillometry · Human Behavior Modeling · Adaptive AI Systems · Neuro-physiological Signal Analysis

EDUCATION

Old Dominion University, Norfolk, VA — *Ph.D., Computer Science* Jan 2019 – Aug 2024

GPA: 4.0/4.0 | Dissertation: “*RAEMAP: Real-Time Advanced Eye Movements Analysis Pipeline*” | *Lee Entsminger Outstanding Dissertation Award (2024-2025)*; *CS Outstanding Researcher Award (2019-2020)*

University of Moratuwa, Sri Lanka — *B.Sc., Computer Science and Engineering*

Jan 2015 – Dec 2018

GPA: 3.85/4.20 | First Class Honors

RESEARCH & WORK EXPERIENCE

The University of Texas at Austin — *Bullard Postdoctoral Research Fellow, School of Information* Sep 2024 – Present

Postdoctoral researcher at the Information eXperience (IX) Lab, working on real-time cognitive load detection, eye-tracking, and applied AI in information interaction and adaptive human-AI interaction.

- Developed **RIPA2**, a **real-time cognitive load estimation** method via pupillometry and refined Savitzky-Golay filtering, outperforming offline pupillary indices across memory and naturalistic tasks. (*JEMR 2025*)
- Built **real-time visual attention classifiers from eye-tracking data**, enabling AI systems to infer perceptual state and intent. (*ETRA 2025*)
- Modeled how confirmation bias and document relevance shift cognitive load via pupillometry, informing adaptive AI-driven information retrieval. (*CHIIR 2025*)
- Quantified cognitive load effects of working memory capacity and task complexity via pupillometry, informing AI systems that adapt to individual user state. (*CHIIR 2026*)
- Identified gaze-based behavioral markers of misinformation susceptibility during credibility assessment, enabling AI models to detect human reasoning patterns from eye-tracking. (*ETRA 2026*)
- Ongoing*: Studying attention and cognitive load during LLM-assisted learning via eye-tracking, informing real-time adaptive LLM systems.

Old Dominion University — *Graduate Research & Teaching Assistant, Dept. of CS*

Jan 2019 – Aug 2024

- Built **RAEMAP**, a real-time human perception pipeline (fixations, saccades, entropy, ambient/focal, cognitive load) with WebSocket streaming and live visualizations. (*Ph.D. dissertation 2024 [Lee Entsminger Outstanding Dissertation Award]*; *ETRA 2022 [Best Doctoral Symposium]*)
- Achieved **first real-time cognitive load discrimination via pupillometry** by redesigning LHIPA into **RIPA** (*KES 2022*)
- Applied ML** (tree-based classifiers, GCN) to model human behavior from eye movements: 91% ADHD prediction accuracy. (*IEEE IRI 2019*; *Book Chapter 2019*)
- Applied computer vision AI (**YOLO**, **Mask R-CNN**) to automate dynamic AOI detection in video stimuli for real-time gaze analysis. (*IEEE IRI 2020*; *IJMDEM 2021*)
- Best Poster Award ×2** for uncovering structured reading patterns via eye-tracking. (*JCDL 2020, 2021*)
- Contributed to **377+** cited review of eye movement & pupil measures. (*Frontiers CS 2022*)
- Teaching Assistant/Instructor**: Introduction to Python (*Su24*); App Development for Smart Devices (*Sp20*); Computer Architecture (*Fa20*); Database (*Fa20*); Problem Solving & Programming II (*Sp21*);

Los Alamos National Laboratory — *Summer Graduate Research Intern (twice)*

Jun–Aug 2021, Jun–Aug 2022

- Built a production Flask + GROBID pipeline for real-time extraction of structured metadata and full text from scholarly PDFs, automating RASSTI review form population, demonstrating applied NLP/document AI in a high-stakes scientific environment. (*LANL Tech. Report, 2022*)

UStocktrade (Pvt) Ltd, Sri Lanka — *Software Engineering Intern*

Jun – Dec 2017

- Architected and delivered **MSstack**, a production-ready microservices framework with abstracted lifecycle management, data partitioning, and auto-scaling — demonstrating systems engineering skills applicable to scalable AI inference infrastructure. (*ICTer 2018*)

- Built an **OpenAPI/Swagger + Mustache** code-generation tool in Java to auto-produce and continuously update API documentation on every code change, improving developer workflow automation.

SELECTED PUBLICATIONS [[Google Scholar](#), 658+ citations, h-index: 11]

- [Jayawardena](#), et al. "Measuring Mental Effort in Real Time Using Pupillometry." *Journal of Eye Movement Research* 18(6), 2025.
- [Jayawardena](#), Zhang, Gwizdka. "Phase-Dependent Scanpath Dynamics in Map-Based Tasks." *ETRA 2026*.
- [Jayawardena](#), Shi, Gwizdka. "Effects of Working Memory Capacity and Search Task Complexity on Cognitive Load." *CHIIR 2026*.
- Shi, [Jayawardena](#), Gwizdka. "Pupillometric Analysis of Cognitive Load in Relation to Relevance and Confirmation Bias." *CHIIR 2025*.
- [Jayawardena](#), et al. "A Real-Time Approach to Capture Ambient and Focal Attention in Visual Search." *ETRA 2025*.
- [Jayawardena](#), et al. "Toward a Real-Time Index of Pupillary Activity as an Indicator of Cognitive Load." *KES 2022 – Procedia CS*.
- Mahanama, ..., [Jayawardena](#), et al. "Eye Movement and Pupil Measures: A Review." *Frontiers in Computer Science*, 2022. [377 citations]
- Mahanama, [Jayawardena](#), et al. "Analyzing Unconstrained Reading Patterns of Digital Documents." *JCDL 2021*. [Best Poster]
- [Jayawardena](#), Jayarathna, Wu. "Analyzing Reading Patterns using Eye Tracking." *JCDL 2020*. [Best Poster]
- [Jayawardena](#), Michalek, Duchowski, Jayarathna. "Audiovisual SIN Performance of Young Adults with ADHD." *ETRA 2020*.
- De Silva, ..., [Jayawardena](#), et al. "A Rule-Based System for ADHD Identification." *MERCon 2019*. [Best Paper]

LEADERSHIP, SERVICE & AWARDS

Awards & Honors

- [2024] Lee Entsminger Outstanding Dissertation Award - *Old Dominion University, Norfolk, VA*
- [2022] Best Doctoral Symposium Presentation Award - *ETRA 2022*
- [2021] Best Poster Award - *ACM/IEEE JCDL 2021*
- [2020] Best Poster Award - *ACM/IEEE JCDL 2020*
- [2019] Best Paper Award - *MERCon 2019, IEEE*
- [2019–21] Dominion Scholar; CS Outstanding Researcher Award; Levinstein Scholarship - *Old Dominion University*
- [2015–18] Dean's List (Semesters 1, 6, 7, 8) - *University of Moratuwa, Sri Lanka*

Professional Service

- [2018–2023] **Leadership** - President of ACM Student Chapter-Women, Old Dominion University 2022-23; President of ACM Student Chapter, University of Moratuwa 2018
- [Ongoing] **Reviewer /Editor** - ACM CHI, ETRA, CHIIR, OzCHI, IUI, JCDL, CIKM, AH, Frontiers in Psychology (Cognition), Acta Psychologica, Frontiers in Virtual Reality, Interacting with Computers - Oxford Academic
- [2019–present] **Undergraduate /graduate Research Mentor** - Supervised students on eye-tracking & gaze analysis projects at Computer Science Department, Old Dominion University; IX Lab, School of Information, The University of Texas at Austin; and NSF Research Experiences for Undergraduates (REU 2023)

SKILLS

Programming	Python, Java, R, JavaScript, SQL, PHP, C#, HTML/CSS
AI/ML	PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace, NumPy, Pandas, SciPy, Matplotlib, GCN, Random Forest, XGBoost, YOLO, Mask R-CNN, OpenCV, Signal processing
Applied AI	Conversational AI, LLM assisted interaction design, Natural Language Processing, Structured text extraction from PDFs
Eye Tracking	Real-time gaze analysis (fixations, saccades, microsaccades, scanpaths), pupillometry, visual attention classification, cognitive load, Tobii, Pupil Labs, LSL, iMotions
Tools	Git, Jupyter, LaTeX
Research Methods	Controlled experiment design, within/between-subjects studies, IRB protocols, mixed-effects models, ANOVA, Qualtrics, User interviews, Surveys